1 Get Real

Geoff's analytical solution to the angle extraction problem only takes into account the phase change assuming the amplitude does not change. Which is reasonable if, when going from the parallel to converging beam, you only expect a small change in transmission from the Fresnel equations and attenuation from the material. However the position of the focus also changes which will mean the electric filed of the THz radiation on the detector will also change, this does not affect the phase but it does change the amplitude. In the converging beam document by Geoff he gives the following for the small angle solution to find the full angle θ_0
For a thin cone

$$T_{rat} = e^{-\frac{1}{2}ik\alpha\theta_0^2} \tag{1}$$

For a Guassian

$$T_{rat} = \frac{1}{1 + \frac{1}{2}ik\alpha\theta_0^2} \tag{2}$$

where $\alpha = D\left(1 - \frac{1}{n}\right)$. These equations are the same as each other for small angles. The T_{rat} is also the ratio of the FFTs of the focus and the collimated data.

$$T_{rat} = \frac{T_{conv}}{T_{col}} = \frac{A_{conv}}{A_{col}} e^{-i\Delta\phi} \tag{3}$$

Where $\Delta\phi$ is the difference in phase between the two. We can now see that equation 1 should really be

$$T_{rat} = \frac{A_{conv}}{A_{col}} e^{-\frac{1}{2}ik\alpha\theta_0^2} \tag{4}$$

I have changed the code in the extraction program so i also calculate θ with this.

$$\theta = \sqrt{\frac{c \ln\left(\frac{|T_{rat}|}{T_{rat}}\right)}{i\pi f \alpha}} \tag{5}$$

So here is what we get for the real and imaginary part of θ when we do this compared with what happens when we do not. As you can see this made only a small change to the real part while setting the imaginary part to zero apart from when there is low SNR or we hit the absorption.

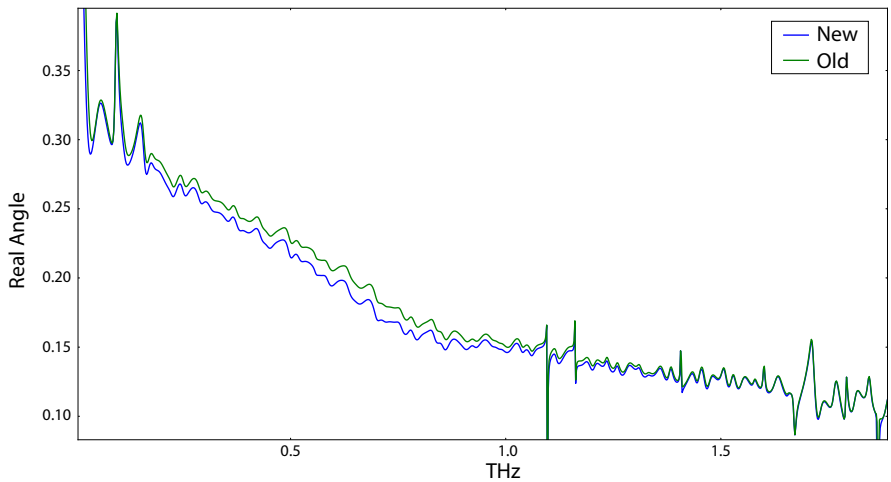

Figure 1: Real Angle

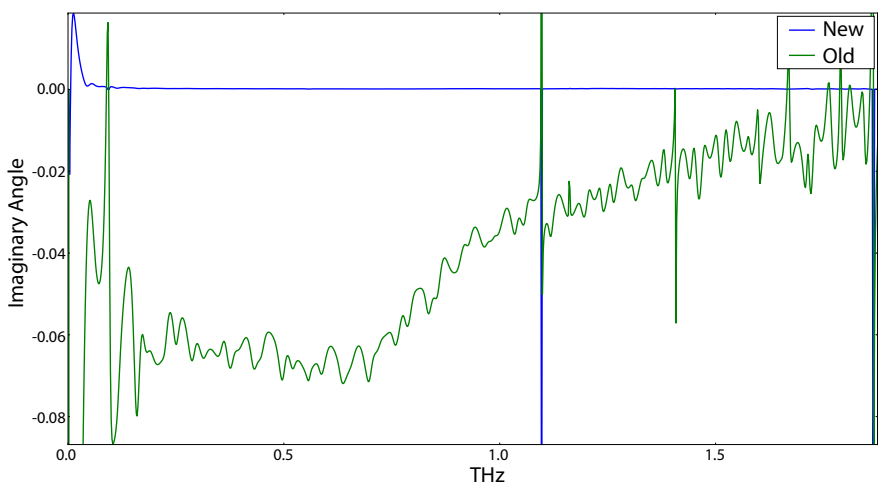

Figure 2: Imaginary Angle